

Instructions:

- **Due March 31, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW07_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework develops the ability to analyze and design feedback control systems using transfer-function models in the basic servo loop architecture. Students will determine stability and steady-state behavior using the transmissibility transfer function and the Final Value Theorem. Emphasis is placed on relating controller gains to closed-loop pole locations and interpreting how these poles govern time-domain performance metrics such as rise time, peak time, overshoot, and settling time. Students will also design proportional and PI controllers by identifying regions of stabilizing gains and mapping performance specifications to regions in the complex plane and gain space. MATLAB and Simulink are used to visualize system dynamics and verify analytical predictions.

Problem 1. Consider the first-order transfer function

$$G(s) = \frac{1}{s - 2} \quad (1)$$

controlled by a proportional controller $G_c(s) = K_p$, where $K_p \in \mathbb{R}$, in a basic servo loop.

1. Determine the values of K_p that render the closed-loop system asymptotically stable. Recall that the closed-loop system is asymptotically stable if the transmissibility transfer function

$$T(s) \triangleq G_{yr}(s) = \frac{G(s)G_c(s)}{1 + G(s)G_c(s)} \quad (2)$$

is asymptotically stable.

2. Let $r(t) = 1(t)$ and choose a stabilizing value of K_p .
 - (a) Determine the closed-loop pole location and the corresponding time constant τ .
 - (b) Using standard first-order formulas, determine the rise time T_r and settling time T_s . Assume the rise time corresponds to the 10%–90% definition and the settling time corresponds to the 2% criterion.
 - (c) Using the Final Value Theorem (FVT), determine the asymptotic value of the output $y(t)$.
 - (d) Implement the closed-loop system in Simulink and simulate the response for $0 \leq t \leq 10$. Plot the output $y(t)$ and verify your analytical predictions.
3. Determine the smallest value of K_p such that the rise time satisfies $T_r = 1$ second. For this value of K_p , repeat parts (a)–(d) above.

Problem 2. Consider the first-order transfer function

$$G(s) = \frac{1}{s - 2} \quad (3)$$

controlled by a PI controller

$$G_c(s) = K_p + \frac{K_i}{s}, \quad (4)$$

where $K_p, K_i \in \mathbb{R}$, in a basic servo loop.

1. Determine the values of K_p and K_i that render the closed-loop system asymptotically stable. Sketch the region of stabilizing gains in the $K_p - K_i$ plane.
2. Let $r(t) = 1(t)$ and choose stabilizing values of K_p and K_i .
 - (a) Determine the natural frequency ω_n and damping ratio ζ of the closed-loop system. Then, determine the peak time T_p , percent overshoot, and settling time T_s . Ignore numerator dynamics when computing these quantities.
 - (b) Based on the value of ζ , determine whether the closed-loop system is underdamped, critically damped, or overdamped.
 - (c) Using the Final Value Theorem (FVT), determine the asymptotic value of the output $y(t)$.
 - (d) Implement the closed-loop system in Simulink and simulate the response for $0 \leq t \leq 10$. Plot the output $y(t)$ and verify your analytical predictions.
 - (e) Let the desired peak time satisfy $T_p \in (1, 2)$ seconds and the desired settling time satisfy $T_s \in (3, 4)$ seconds. Sketch the region in the complex plane where the closed-loop poles must lie so that the response satisfies these specifications. Sketch the corresponding region in the $K_p - K_i$ plane. Select values of K_p and K_i from this region and repeat parts (a)–(d).